Cutaneous lymphomas

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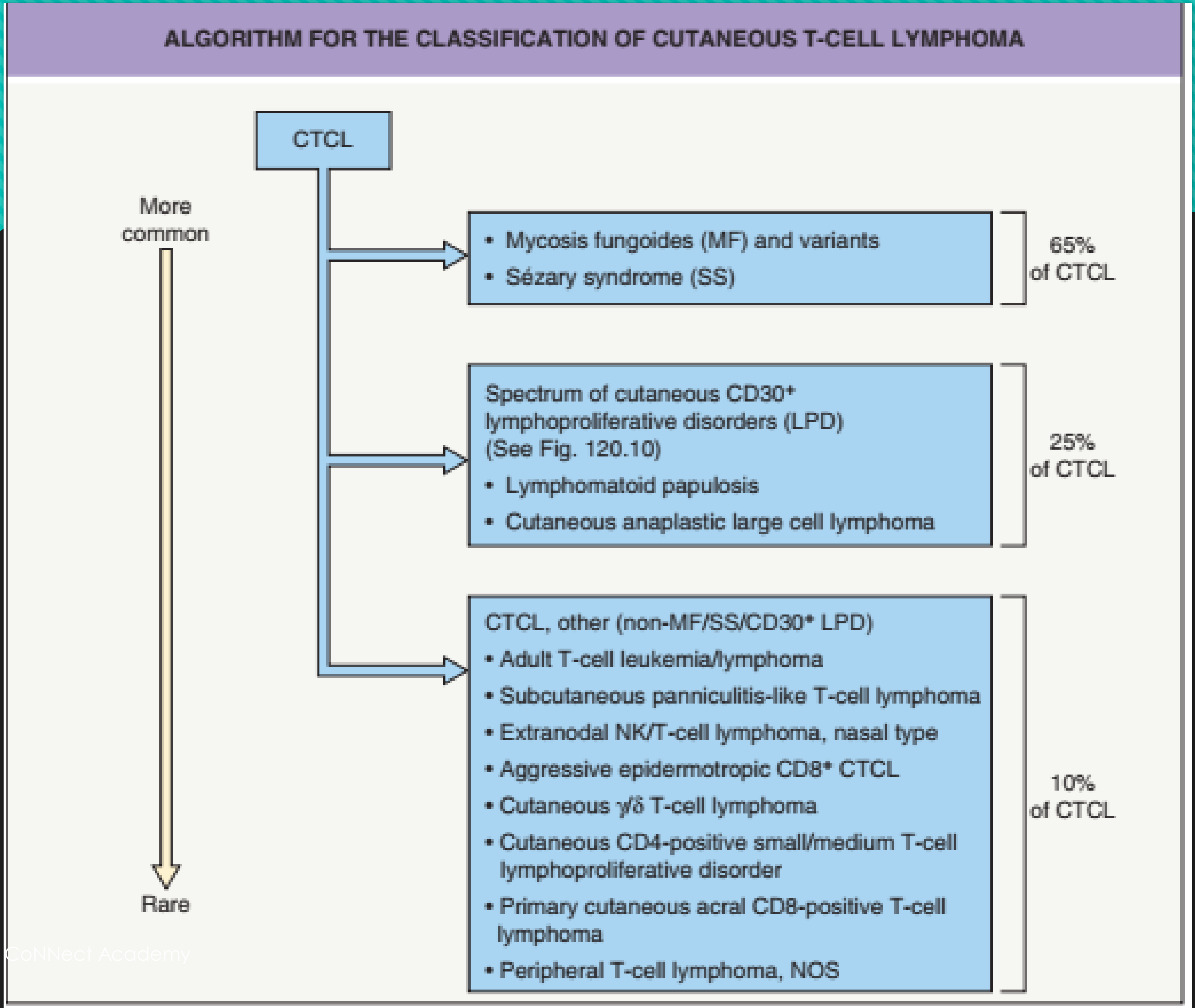
Consultant dermatologist

- Lymphomas
  - Extracutaneous Lymphoma
    - 2ry CLS
  - Cutaneous lymphomas
    - 1ry cls
    - 2ry cls

- 1ry cutaneouse t cell lymphoma
  - T cell lymphoma
    - Indolent
    - aggressive
  - B cell lymphoma
    - Indolent
    - Aggressive

WHO-EORTC CLASSIFICATION FOR CUTANEOUS T-CELL LYMPHOMAS		
WHO-EORTC classification	Frequency (%)*	5-year survival rate (%)*
Indolent clinical behavior		
Mycosis fungoides (MF)	54	88
Mycosis fungoides variants		
• Folliculotropic MF	6	80
• Pagetoid reticulosis	1	100
• Granulomatous slack skin	<1	100
Primary cutaneous CD30-positive lymphoproliferative disorders		
• Primary cutaneous anaplastic large cell lymphoma (C-ALCL)	10	95
• Lymphomatoid papulosis (LyP)	16	100
Subcutaneous panniculitis-like T-cell lymphoma (SPTCL)	1	82
Primary cutaneous CD4-positive small/medium T-cell lymphoproliferative disorder¶	3	75#
Primary cutaneous acral CD8-positive T-cell lymphoma¶	<1	100
Aggressive clinical behavior		
Sézary syndrome (SS)	4	24
Adult T-cell leukemia/lymphoma (ATLL)	NDA	NDA
Extranodal NK/T-cell lymphoma, nasal type	1	<5
Primary cutaneous CD8-positive aggressive epidermotropic cytotoxic T-cell lymphoma¶	<1	18
Primary cutaneous gamma/delta T-cell lymphoma (PCGD-TCL)	1	<5
Primary cutaneous peripheral T-cell lymphoma (PTCL), NOS	3	16


ALGORITHM FOR THE CLASSIFICATION OF CUTANEOUS T-CELL LYMPHOMA
CTCL
More common
Rare
• Mycosis fungoides (MF) and variants
• Sézary syndrome (SS)
65% of CTCL
Spectrum of cutaneous CD30+ lymphoproliferative disorders (LPD) (See Fig. 120.10)
• Lymphomatoid papulosis
• Cutaneous anaplastic large cell lymphoma
25% of CTCL
CTCL, other (non-MF/SS/CD30+ LPD)
• Adult T-cell leukemia/lymphoma
• Subcutaneous panniculitis-like T-cell lymphoma
• Extranodal NK/T-cell lymphoma, nasal type
• Aggressive epidermotropic CD8+ CTCL
• Cutaneous γ/δ T-cell lymphoma
• Cutaneous CD4-positive small/medium T-cell lymphoproliferative disorder
• Primary cutaneous acral CD8-positive T-cell lymphoma
• Peripheral T-cell lymphoma, NOS
10% of CTCL

Mycosis fungoid

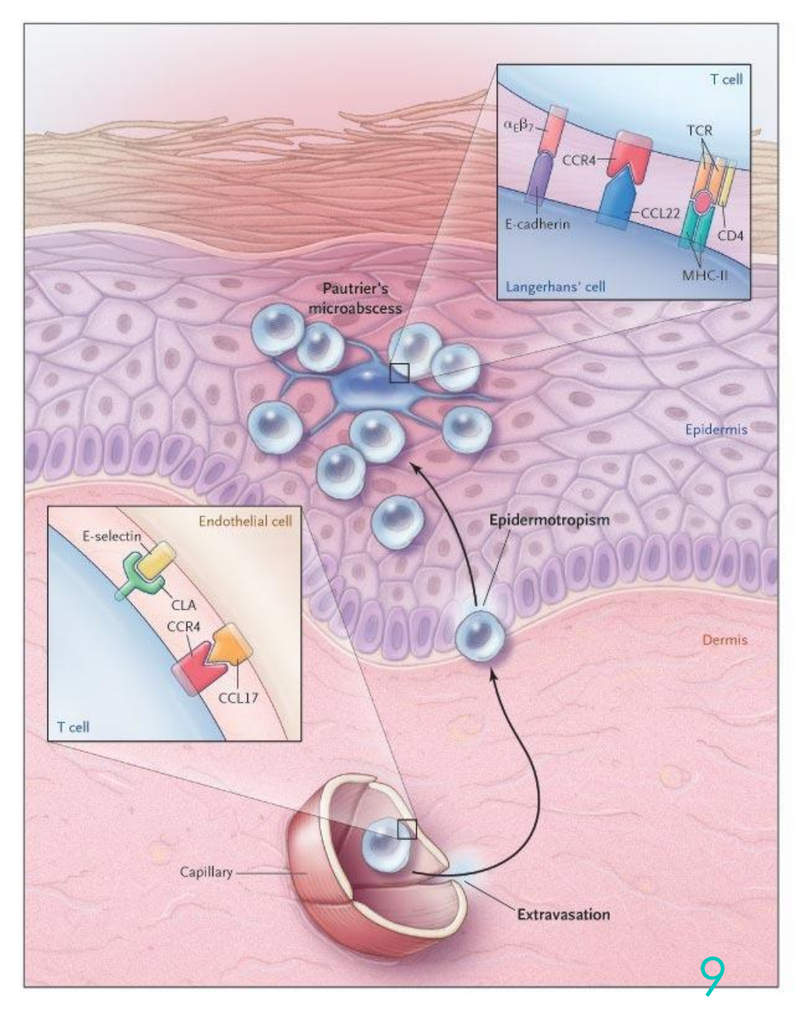
Rare
Most common type of pctcl
Older adult 55- 60 male

aetiopathogenesis

- Enviromental
- genatic
- immunological

aetiopathogensis

- **Genatic**
  - Accumulation of genatic abnormalties >>>>>
  - Colonal malformation , malignant transformation of lymphocyte



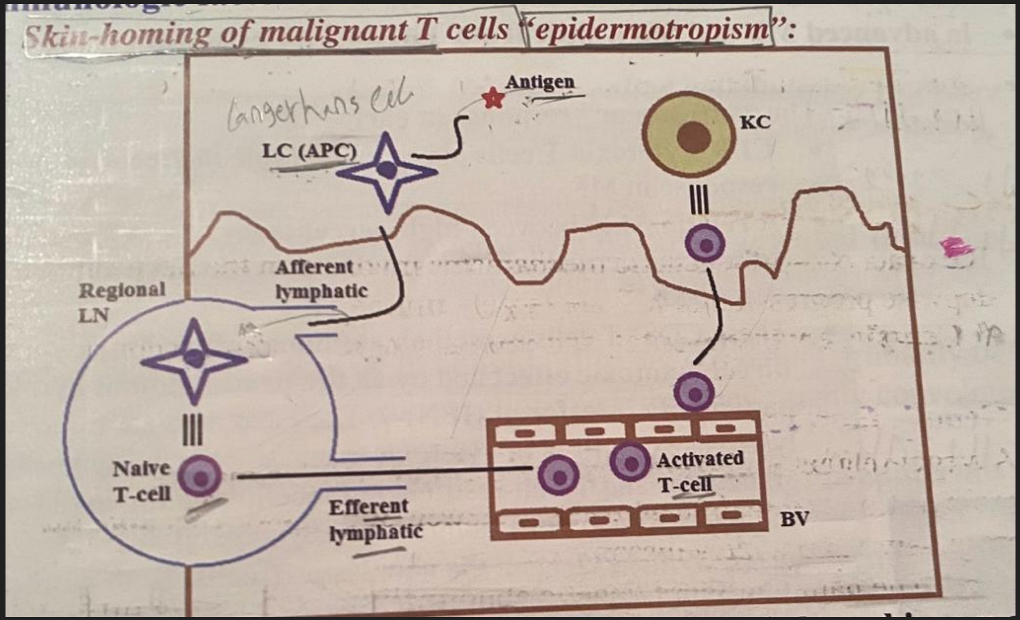
Environmental

- Persistent antigenic stimulation >>>>>>
- Chronic inflammation >>>>>uncontrolled clonal expantion accumulation

1. Staphaureus
2. HTLV 1
3. EBV
4. Immunosuppresion
5. Occuptional

Immunological factor

- skin homming of malignant Tcell
  - Epidermotropisim
  - High expression of adhesion molecules , chemokin
  - ccR4 , ccR 10 , cxcr3
  - Bind to crosponding ligand on kcs , lcs , endothelial cell



Immunophentyping

- Memory T helper
- **CD3-CD4 +ve**
- **CD7-CD8   -ve**
- CD7 –ve >>>>> sensitive specific finding
- CD4 :CD8 = > 6

Clinical picture

3 successive stages progress slowly over years

Start with nin specific eczematous or ps form skin lesion

Patch stag

Plaque stag

tumor stag

Patch stage

- Mildly pruritic erythamtus scaly patch with atrophy
- Covered area (bathing suit
- Varient
  - Poikeloderma vascular atrophicans
  - Hypopigmented (dark skin

Histeopathology :

Superficial band like >>>>lichenoid infiltrate >>>>atypical lymphocyte >>>epidermotropisim

Single cells surrounded by vaculated hallow

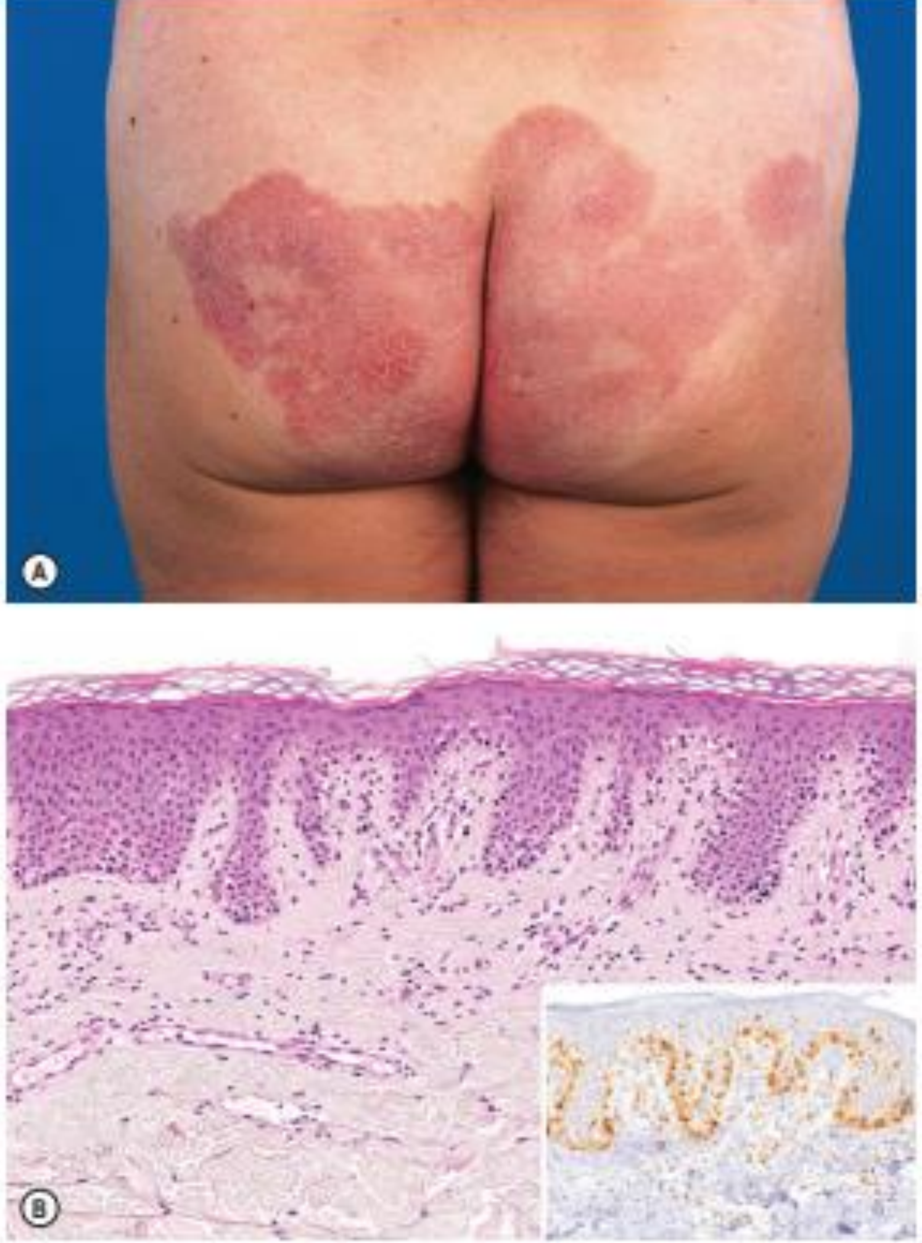


Fig. 120.2 Mycosis fungoides, limited patch/plaque stage disease (stage 1A). **A** Patches on the buttocks involving less than 10% of the skin surface. **B** Few atypical T cells in the basal layer of the epidermis. Immunohistochemical staining for CD3 more clearly demonstrates many atypical lymphocytes in a characteristic linear configuration along the epidermal basal layer (inset).

Patch stage

- Infilrated redish brown scaling plaque
- Polycyclic horse sheo shape
- Histeopathology :
  - Increase epidermotropisim(pautrier microabscesses)
  - Intraepidermal nest of atypical cells
  - Increase dermal infeltirate
  - Compact hyperkeratosis , patchy parakeratosis

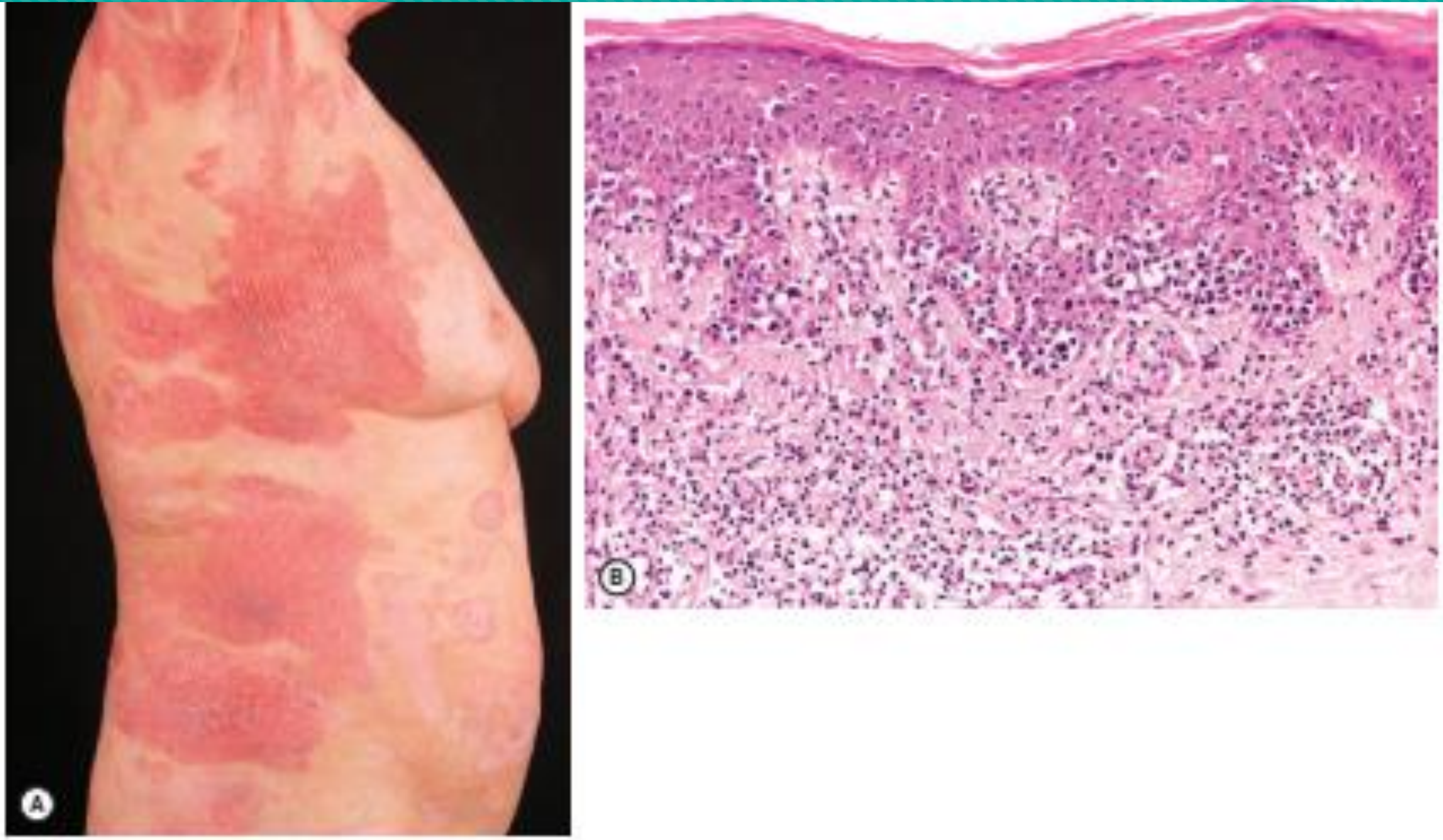


Fig. 120.4 Mycosis fungoides, generalized patch/plaque stage disease (stage 1B). A Extensive patches and plaques with scale involving more than 10% of the skin surface. B Epidermotropism with the formation of small nests of atypical cells (Pautrier microabscesses). The majority of the dermal infiltrate is composed of small reactive lymphocytes.
A, Courtesy, Lorenzo Cerroni, MD.

Tumor stage

  - Compination of patch , plaque , tumor nodule , ulceration
- Histopathology :
  - No epidermotropisim
  - Increase tumor cell number
  - In all dermis s.c tissue

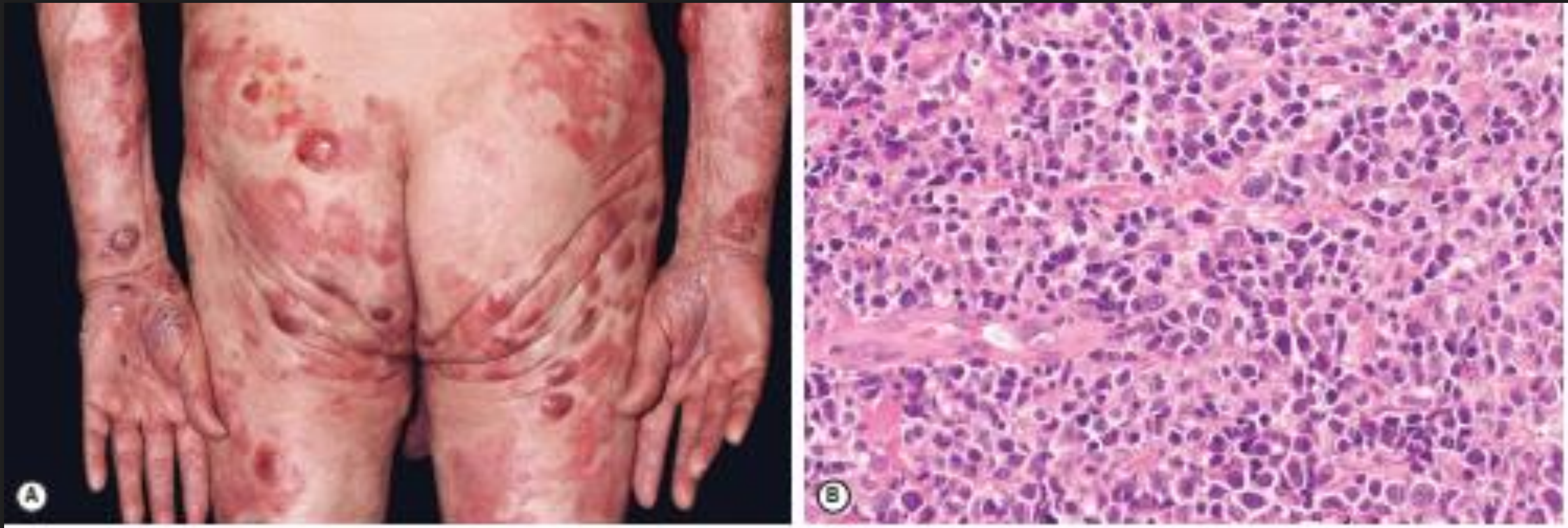


Fig. 120.5 Mycosis fungoides, tumor stage. A Multiple skin tumors in combination with typical patches and plaques. B. Diffuse dermal infiltrates of medium-sized to large neoplastic T cells.

N.B

- Aatypical lymphocyte
  - Larg cell with irregular nuclei
  - Multilobed hyperconvoluted nuclei

Staging

TNMB CLASSIFICATION OF MYCOSIS FUNGOIDES AND SÉZARY SYNDROME	
T (skin)	
T_1	Limited patch/plaque (involving <10% of total skin surface)
T_2	Generalized patch/plaque (involving ≥10% of total skin surface)
T_3	Tumor(s)
T_4	Erythroderma
N (lymph node)	
N_0	No enlarged lymph nodes
N_1	Enlarged lymph nodes, histologically uninvolved
N_2	Enlarged lymph nodes, histologically involved (nodal architecture uneffaced)
N_3	Enlarged lymph nodes, histologically involved (nodal architecture (partially) effaced)
M (viscera)	
M_0	No visceral involvement
M_1	Visceral involvement
B (blood)	
B_0	No circulating atypical (Sézary) cells (or <5% of lymphocytes)
B_1	Low blood tumor burden (≥5% of lymphocytes are Sézary cells, but not B_2)
B_2	High blood tumor burden (≥1000/mcl Sézary cells + positive clone)

CLINICAL STAGING SYSTEM FOR MYCOSIS FUNGOIDES AND SÉZARY SYNDROME				
Clinical stage				
IA	T_1	N_0	M_0	B_{0-1}
IB	T_2	N_0	M_0	B_{0-1}
IIA	T_{1-2}	N_{1-2}	M_0	B_{0-1}
IIB	T_3	N_{0-1}	M_0	B_{0-1}
III	T_4	N_{0-2}	M_0	B_{0-1}
IVA$_1$	T_{1-4}	N_{0-2}	M_0	B_2
IVA$_2$	T_{1-4}	N_3	M_0	B_{0-2}
IVB	T_{1-4}	N_{0-3}	M_1	B_{0-2}

Treatment :

Early stag

1. **Topical steroid**
2. **Topical cytoxic**
   1. Nitrogen mustard
3. **Phototherapy**
   1. Puva :standerd
   2. Nbuvb
   3. Extracorporial photopheresis
4. **Radiotherapy**
5. **Interferon**
6. **Acitretin , bexrotene**
7. **New therapy** :
   1. Denileukin
   2. Dititox

Late stag

Systemic multiagent chemotherapy

- 6 cycle of chop
  1. Cyclophosphamid
  2. Hydroxydonomycin
  3. Oncoven
  4. Prednison
- Allogenic hematopoitic stem cell transplantation

Treatment

TREATMENT OF MYCOSIS FUNGOIDES
Premycotic phase
• Topical or intralesional corticosteroids; narrowband UVB[§] • If former not effective, PUVA[§]
Stage IA–IIA (patches/plaques)
• PUVA[§]; HN2 (see Ch. 129) • Narrowband UVB[§] (if only patches) • Topical corticosteroids or topical retinoids[^] (if only limited patches/thin plaques as second-line therapy) • RT (if single lesion) • TSEB (if generalized thick plaques)
Stage IIB (skin tumors)
• PUVA or HN2 + RT (if only a few tumors) • TSEB (followed by skin-directed therapies) • Relapse: PUVA + IFN-α; PUVA + retinoids (acitretin; oral bexarotene[*]); second-line therapy: denileukin diftitox[*,¶]; HDACi (vorinostat, romidepsin)[*] • Add RT (if persistent tumors); consider second course of TSEB (10–20 Gy)
Stage III (erythroderma)
• ECP; if not effective, add IFN-α • Low-dose chlorambucil and prednisone; low-dose methotrexate • Add skin-directed therapies (PUVA; HN2; RT), if necessary • Second-line therapy: oral bexarotene[*]; denileukin diftitox[*,¶]; HDACi[*]; low-dose alemtuzumab
Stage IV (nodal, visceral involvement)
• Multi-agent chemotherapy (e.g. CHOP) • Biological response modifiers (denileukin diftitox[¶]; IFN-α; oral retinoids) • Add skin-directed therapies (PUVA; HN2), if necessary • Allogeneic hematopoietic stem cell transplantation (utilizing non-myeloablative reduced-intensity conditioning regimens), in selected cases

MF VARIENTS , SUBTYPES

- Folliculotropic mf
- Pagetoid reticulosis
- Granulomatus slack skin

1. FOLLICULOTROPIC MF

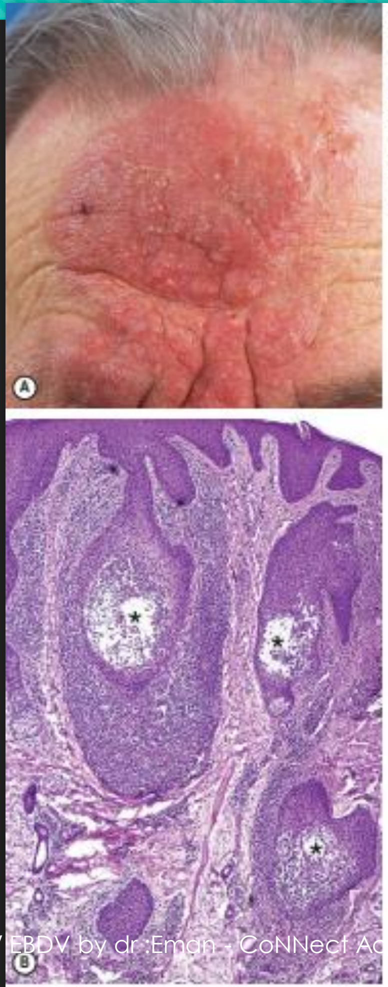


Fig. 120.6 Folliculotropic mycosis fungoides. A Infiltrated plaques on the forehead and eyebrow with concurrent hair loss. B Characteristic perifollicular infiltrates with extensive follicular mucinosis (asterisks). Note absence of epidermotropism.

- Mf associated mucinosis
- Sever pruritic
- Head , neck
- Grouped follicular papules or acneform eruption
- Allopecia mucinuria
- Folliculotropism
- CD3 CD4 +VE
- CD8 CD7 –VE

Not respond to skin directed therapy

eFig. 120.2
Folliculotropic mycosis fungoides. Acneiform lesions and cysts in the face.

Pagetoid reticulosis

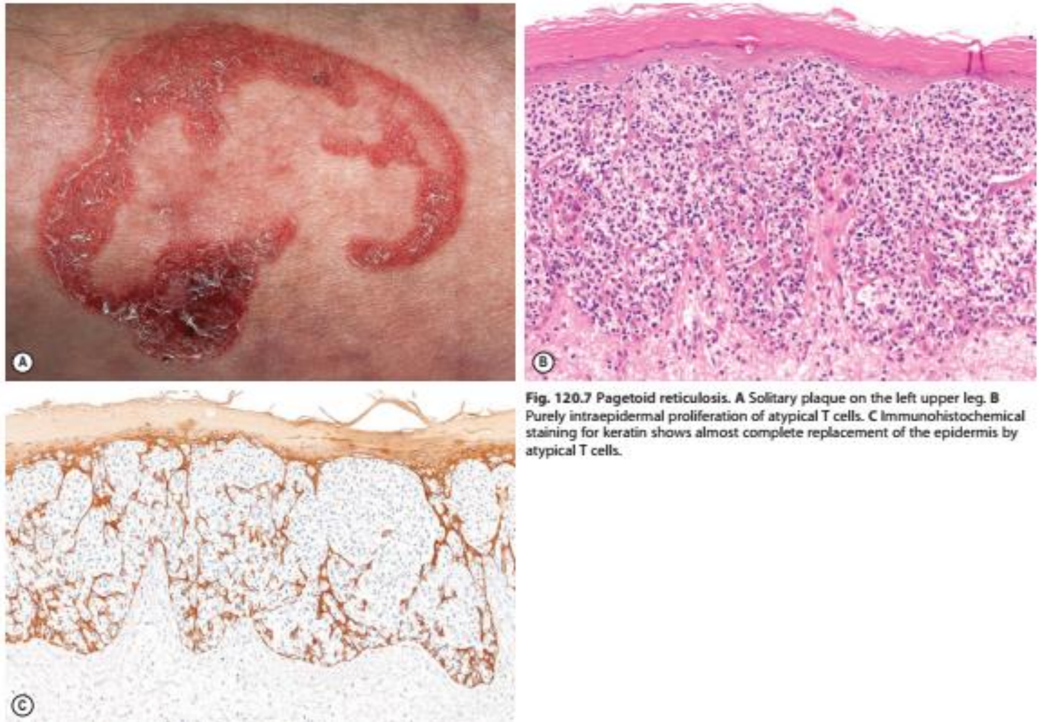


Fig. 120.7 Pagetoid reticulosis. A Solitary plaque on the left upper leg. B Purely intraepidermal proliferation of atypical T cells. C Immunohistochemical staining for keratin shows almost complete replacement of the epidermis by atypical T cells.

- Woringer kolopp
- Unilesional mf
- Solitary slowly progressive psoriasiform or hyperkeratotic patch , plaque
- Types :
  - Localized :woringer kolopp
  - Dissimenated :ketron goodman
- Massive infiltration >>strictly epidermal localization
- Good prognosis

Granulomatus slack skin

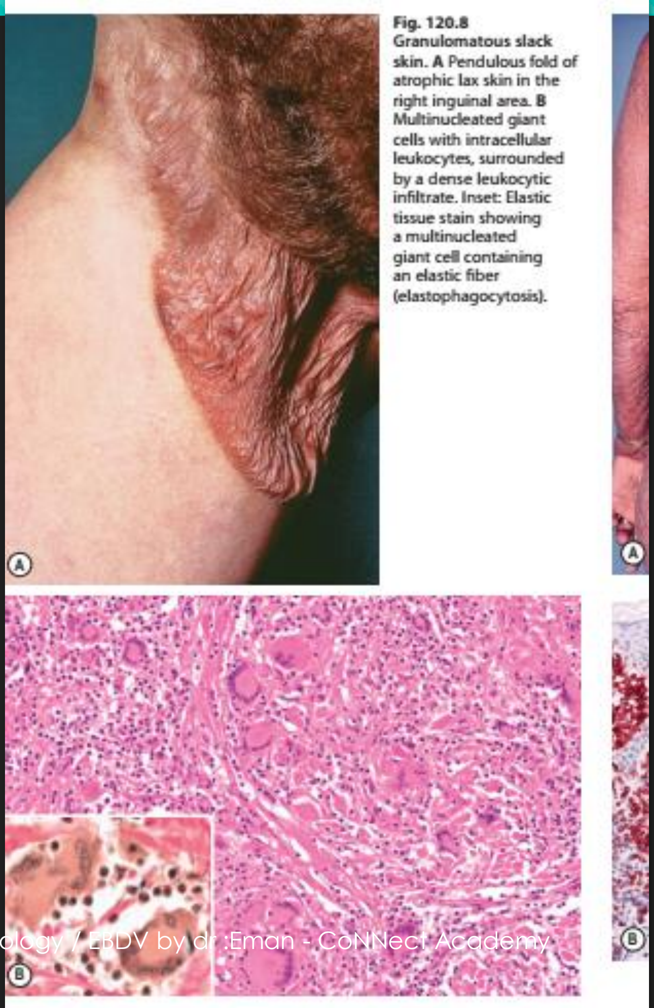


Fig. 120.8 Granulomatous slack skin. A Pendulous fold of atrophic lax skin in the right inguinal area. B Multinucleated giant cells with intracellular leukocytes, surrounded by a dense leukocytic infiltrate. Inset: Elastic tissue stain showing a multinucleated giant cell containing an elastic fiber (elastophagocytosis).

- Extremly rare
- Adolescent , adult men
- Lax skin in axilla , groin
- Third of cases >>>Hodgkin lymphoma
- H.P:
  - Granuloma
  - Atypical t cell
  - elastophagocytolsis

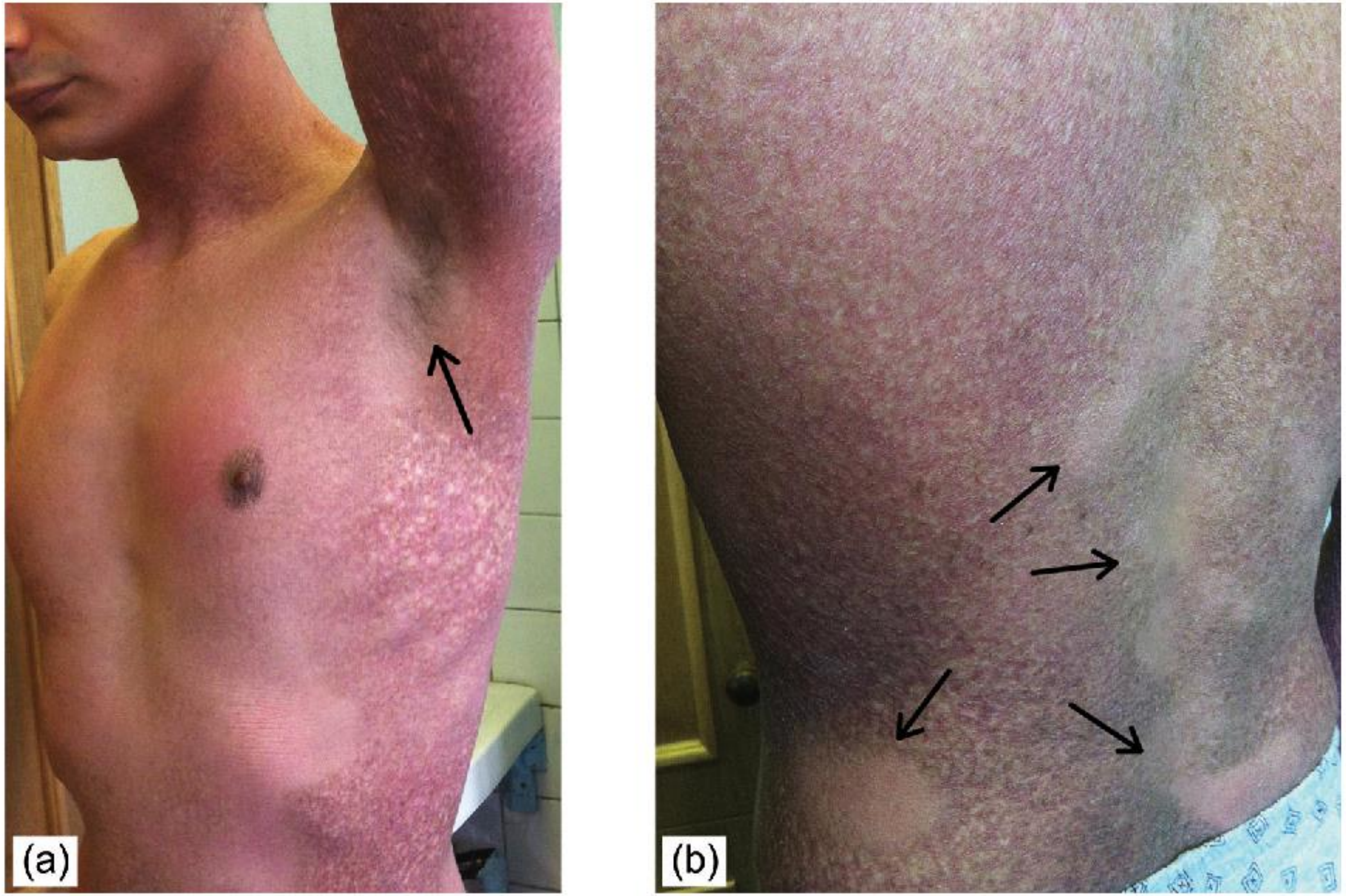
Other varients of mf

Online o

eFig. 120.1 Mycosis fungoides (stage 1B). Generalized hypopigmented lesions as well as pink-colored plaques.

1. Hypopigmented
2. Hyperpigmented
3. Poikilodermatous mf
4. Erythrodermic
5. Vesiculo boiious

lenies any episodes of weight loss. The pa- ill-defined erythematous patches could be obse

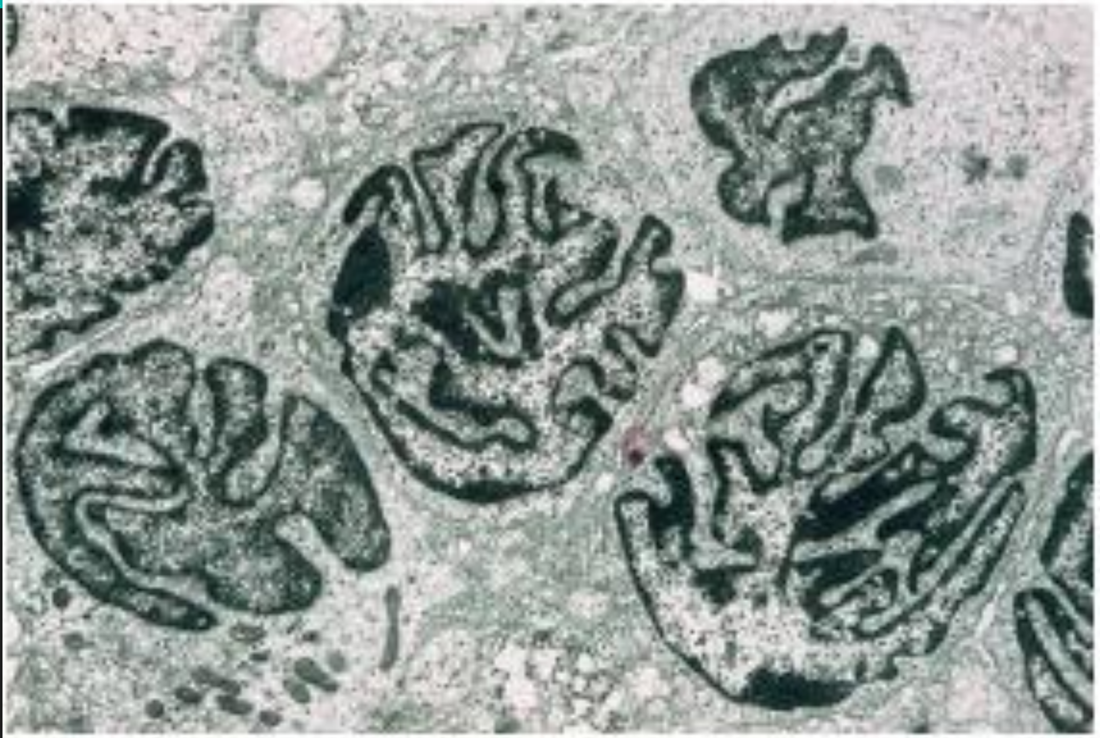


ermatous skin lesions. (a) Spared axillary hair; (b) small islands of uninvolved skin.

Sezary syndrome

Fig. 120.9 Sézary syndrome. A Erythroderma with diffuse scaling. B Strong expression of PD-1 by neoplastic T cells within the dermal infiltrate and Pautrier microabscesses.

- Triad of :
  1. Erythroderma
  2. Generalized lymphadenopathy
  3. Neoplastic t cell in skin , LN ,blood
     - Sezary cells >1000/mcl
  - Intensly pruritic , lymphadenopathy , alopecia , onychodystrophy ,
  - TTT :
    - Extracorborial photopheresis
    - Ifn alpha
    - Chop



eFig. 120.3 Sézary syndrome. Electron photomicrograph of a skin biopsy showing characteristic Sézary cells.

CD30 LYMPHOPROLEFERATIVE DISORDER :

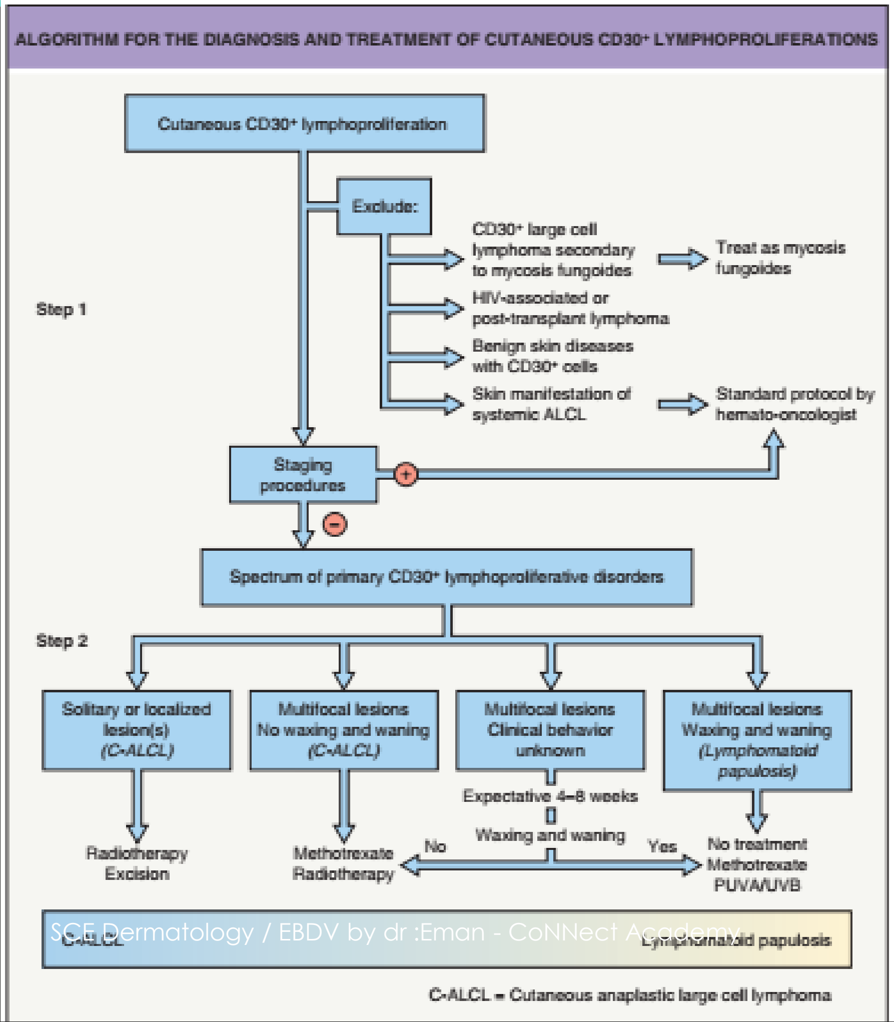


Fig. 120.10 Algorithm for the diagnosis and treatment of primary cutaneous CD30-positive lymphoproliferations. Cases previously designated as regressing atypical histiocytosis, as well as rare cases of primary cutaneous Hodgkin lymphoma with an indolent clinical course, also belong to this spectrum. Of note, there are reports of ALCL developing in patients with breast implants.
From ref. 66.

1. *Lymphomatoid papulosis*
2. *CALCL*

Lymphomatoid papulosis

- Chronic recurrent self healing >>papulonecrotic , papulonodular >>
- with CD 30+
- Viral infecton :HTCLV1 , EBV , HSV1,2 , HHV6
- C/P:
  - Red , brown papules ,nodules >>>with central HGE , necrosis
  - Heal spontaneusly with 3-12 wks recurrent

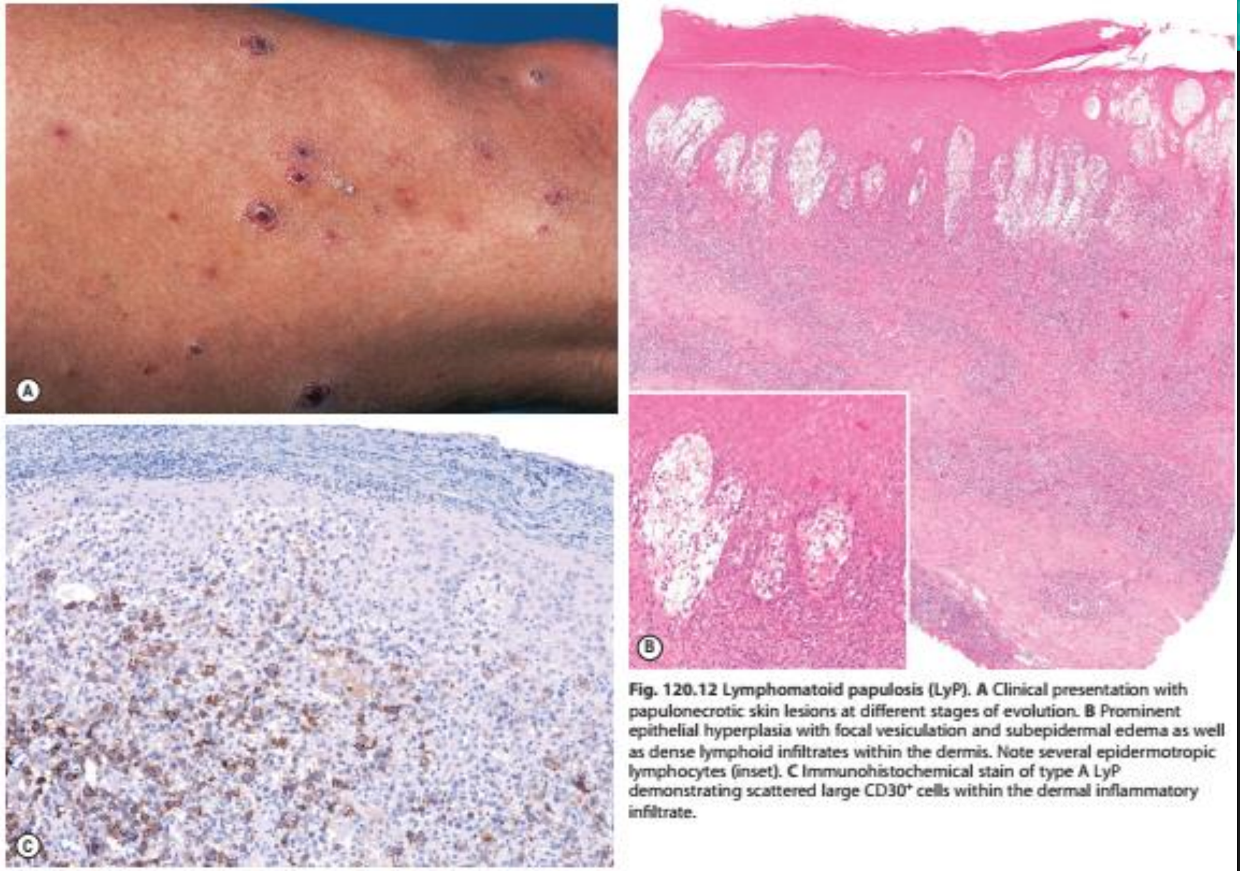


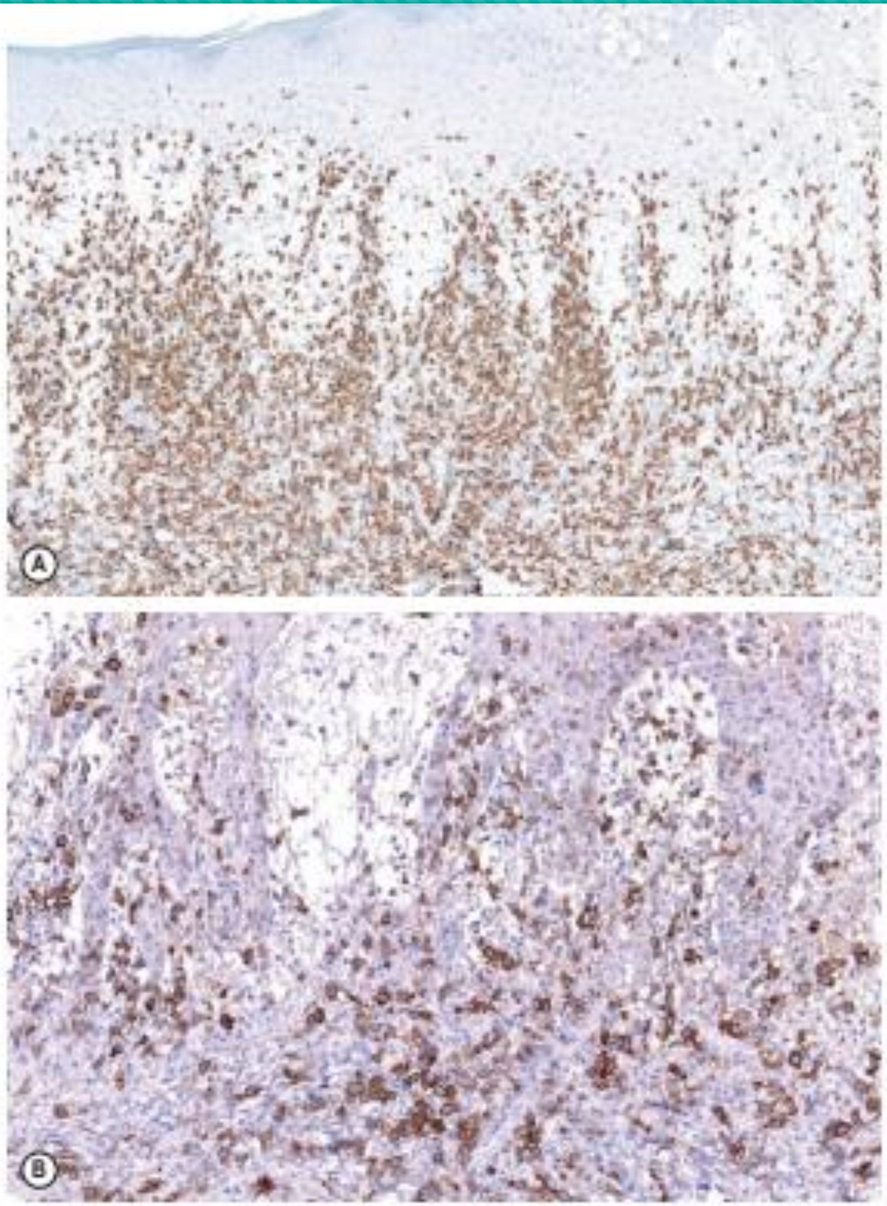
Fig. 120.12 Lymphomatoid papulosis (LyP). A Clinical presentation with papulonecrotic skin lesions at different stages of evolution. B Prominent epithelial hyperplasia with focal vesiculation and subepidermal edema as well as dense lymphoid infiltrates within the dermis. Note several epidermotropic lymphocytes (inset). C Immunohistochemical stain of type A LyP demonstrating scattered large CD30⁺ cells within the dermal inflammatory infiltrate.

- DD
  - PLEVA , PLC , FOLLICULITIS , INSECT BITE , VARICELLA
- TT:
  - Long term follow up in all pts
  - **Low dose methotrexate**
  - Radiotherapy

LYMPHOMATOID PAPULOSIS (LyP) – HISTOLOGIC SUBTYPES AND THEIR DIFFERENTIAL DIAGNOSES		
LyP type	**Characteristic histologic features**	**Primary differential diagnosis**
Type A[66]	Wedge-shaped infiltrate with scattered or small clusters of large, atypical, sometimes multinucleated or Reed–Sternberg-like, $CD30^+$ T cells The T cells are within an extensive inflammatory infiltrate composed of small lymphocytes, histiocytes, neutrophils, and/or eosinophils Neutrophils may be scattered within the epidermis	C-ALCL Tumor stage MF Hodgkin lymphoma
Type B[66]	Epidermotropic, with a superficial perivascular to band-like and sometimes wedge-shaped infiltrate of small to medium-sized, atypical $CD4^+$, $CD30^+$ or $CD30^-$ T cells[^]	Patch/plaque stage MF
Type C[66]	Diffuse infiltrates or large clusters of large $CD30^+$ T cells, with relatively few admixed inflammatory cells	C-ALCL Transformed MF ($CD30^+$)
Type D[103]	Marked epidermotropism of atypical, small to medium-sized, $CD8^+$, $CD30^+$ pleomorphic T cells Necrotic keratinocytes may be present	$CD8^+$ aggressive epidermotropic T-cell lymphoma Pityriasis lichenoides et varioliformis acuta
Type E[104]	Prominent angioinvasion and angiodestruction by medium-sized $CD8^+$, $CD30^+$ pleomorphic T cells Vascular occlusion and/or thrombi; hemorrhage, extensive necrosis, and ulceration	Extranodal NK/T-cell lymphoma
Type F[105]	Perifollicular infiltrates with variable degree of folliculotropism of atypical $CD30^+$ T cells	Folliculotropic MF
DUSP-IRF4 type[106*]	Extensive epidermotropism by weakly $CD30^+$ small to medium-sized T cells with cerebriform nuclei Strongly $CD30^+$ medium-sized to large blast cells in the dermis	Transformed MF C-ALCL

^If proper immunohistochemical method is used, $CD30^-$ cases should be regarded critically.

*Chromosomal rearrangements involving the *DUSP-IRF4* locus on 6p25.3; localized skin lesions in older individuals.



eFig. 120.4 Lymphomatoid papulosis, type D. A Strong expression of CD8.
B Many cells, including intraepidermal ones, express CD30.

COMPARISON OF SUBCUTANEOUS PANNICULITIS-LIKE T-CELL LYMPHOMA AND PRIMARY CUTANEOUS GAMMA/DELTA T-CELL LYMPHOMA WITH SUBCUTANEOUS INVOLVEMENT		
	SPTCL	PCGD-TCL with subcutaneous involvement
Phenotype	α/β T-cell phenotype	γ/δ T-cell phenotype
T-cell receptor	$\beta F1^+$, $TCR\gamma 1^-$	$\beta F1^-$, $TCR\gamma 1^+$
T-cell phenotype	$CD3^+$, $CD4^-$, $CD8^+$	$CD3^+$, $CD4^-$, $CD8^-$
Coexpression of CD56	Absent	Common
Architecture	Subcutaneous	Subcutaneous and epidermal/dermal
Clinical features	Nodules and plaques Rarely ulceration	Nodules and plaques Ulceration common
Hemophagocytic syndrome	Uncommon	Common
Survival (5-year)	>80%	<10%
Treatment	Systemic corticosteroids	Systemic chemotherapy
WHO-EORTC classification (2004)[3] WHO classification (2008)/2016[4]	Subcutaneous panniculitis-like T-cell lymphoma	Primary cutaneous gamma/delta T-cell lymphoma

Gamma ,delta T cell lymphoma

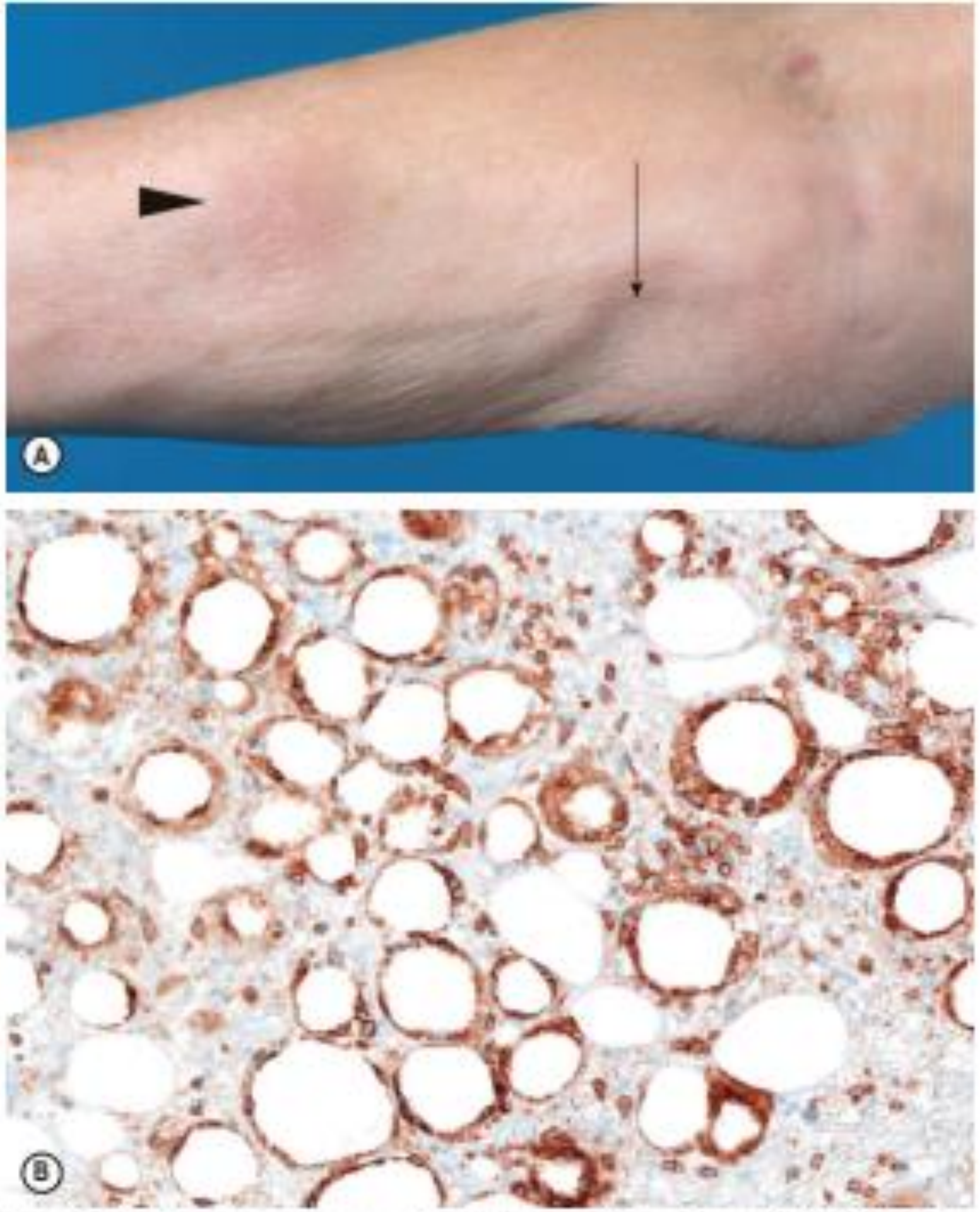


Fig. 120.13 Subcutaneous panniculitis-like T-cell lymphoma (SPTCL). A Pink nodular skin lesions (arrowhead) as well as an area of lipoatrophy (arrow) at the site of a resolved nodule. B Subcutaneous infiltration with rimming of CD8-positive neoplastic T cells around adipocytes.

NK/ T CELL lymphoma nasal type

Fig. 120.14 Nasal NK/T-cell lymphoma. An extensive ulceronecrotic nasal mass, previously referred to as lethal midline granuloma.

- EBV
- ADULT MEN
- V aggressive , ulceration

Fig. 120.16 Primary cutaneous peripheral T-cell lymphoma, NOS. Rapidly growing nodules and tumors. Note: the skin lesions in the left upper corner do not represent patches, but rather deep plaques present for 2

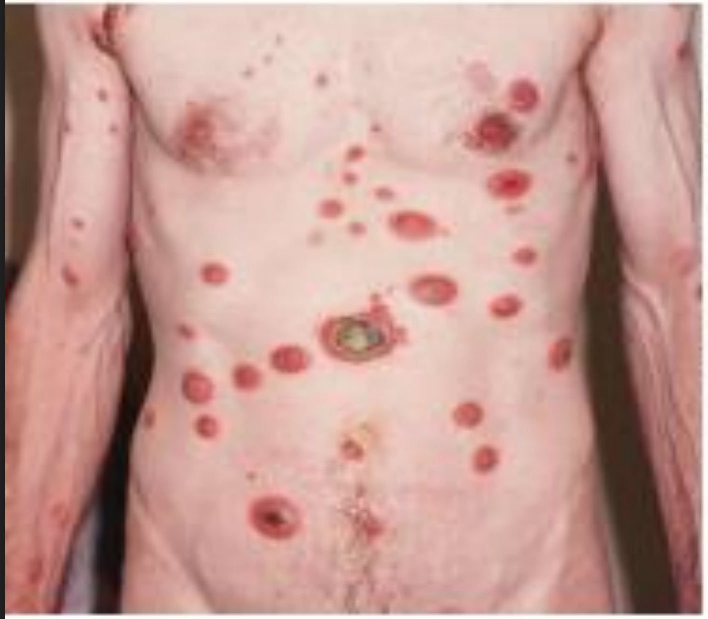


Fig. 120.15 Primary cutaneous CD8-positive aggressive epidermotropic cytotoxic T-cell lymphoma. Generalized skin tumors with central ulceration. Previously, such cases were designated as disseminated pagetoid reticulosis (Ketron–Goodman


Thank
You